Komorebi Space

One realization, seen at four resolutions.

COHERENT VIEWS

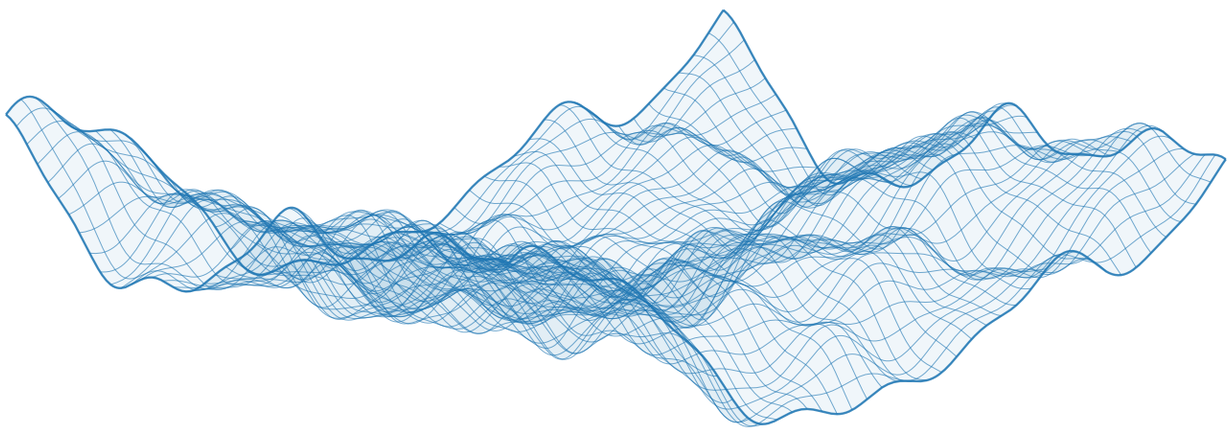
RETAINED DETAIL

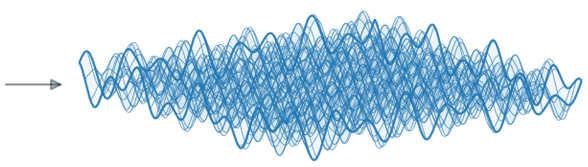
Detail height magnified 2×

x_3

$|k|_\infty \le 8$

$\dim X_3 = 289$





$d_2 = x_3 - J_2 x_2$

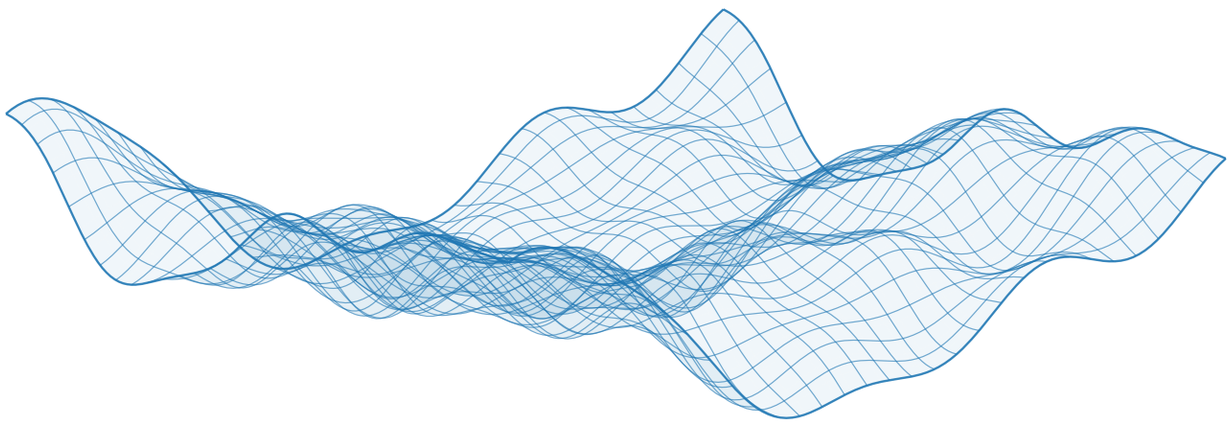
$P_2 d_2 = 0$

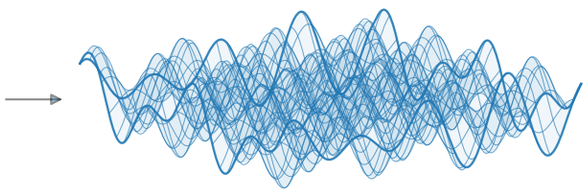
P_2

x_2

$|k|_\infty \le 4$

$\dim X_2 = 81$





$d_1 = x_2 - J_1 x_1$

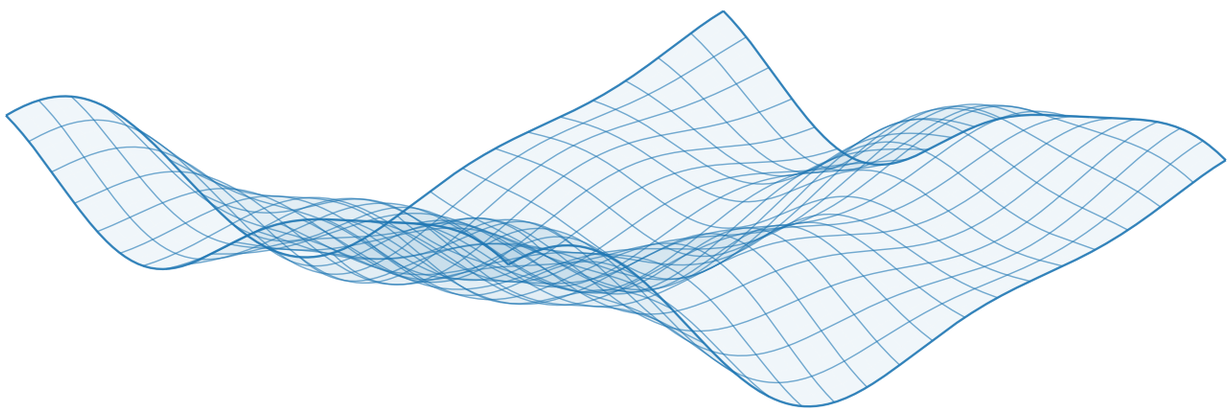
$P_1 d_1 = 0$

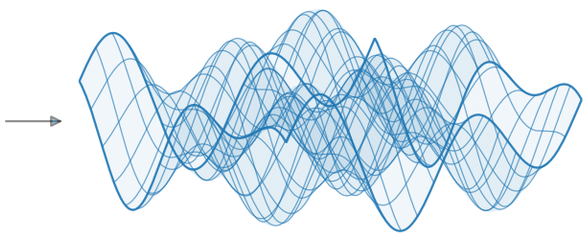
P_1

x_1

$|k|_\infty \le 2$

$\dim X_1 = 25$





$d_0 = x_1 - J_0 x_0$

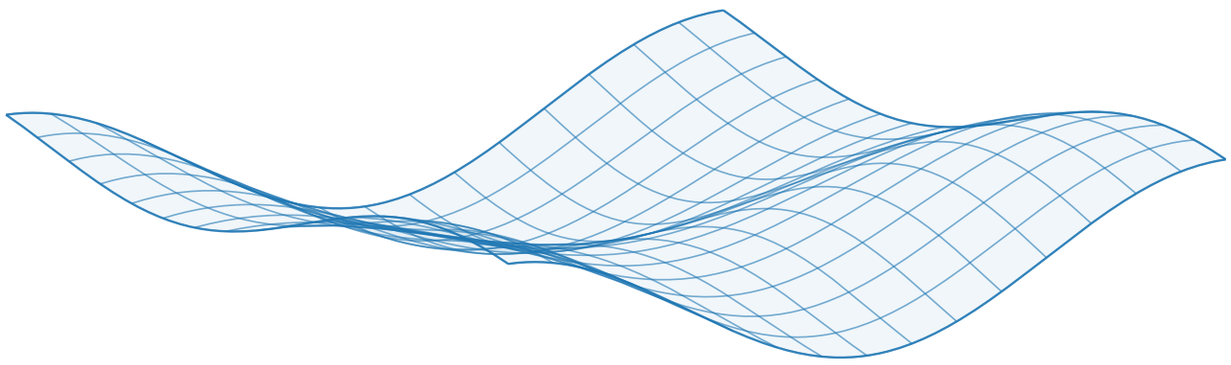
$P_0 d_0 = 0$

P_0

x_0

$|k|_\infty \le 1$

$\dim X_0 = 9$



Detail is information,

not inconsistency.

$P_n x_{n+1} = x_n$

Every restriction agrees exactly.

THE POINT IS THE WHOLE FAMILY

$$\mathcal{K}_3 = \{(x_0, x_1, x_2, x_3) \in \prod_{n=0}^{3} X_n \ : \ P_n x_{n+1} = x_n \quad (n = 0, 1, 2)\}$$

Each sheet is one Fourier function on the square torus, not the entire space X_n. Main sheets share one height scale.

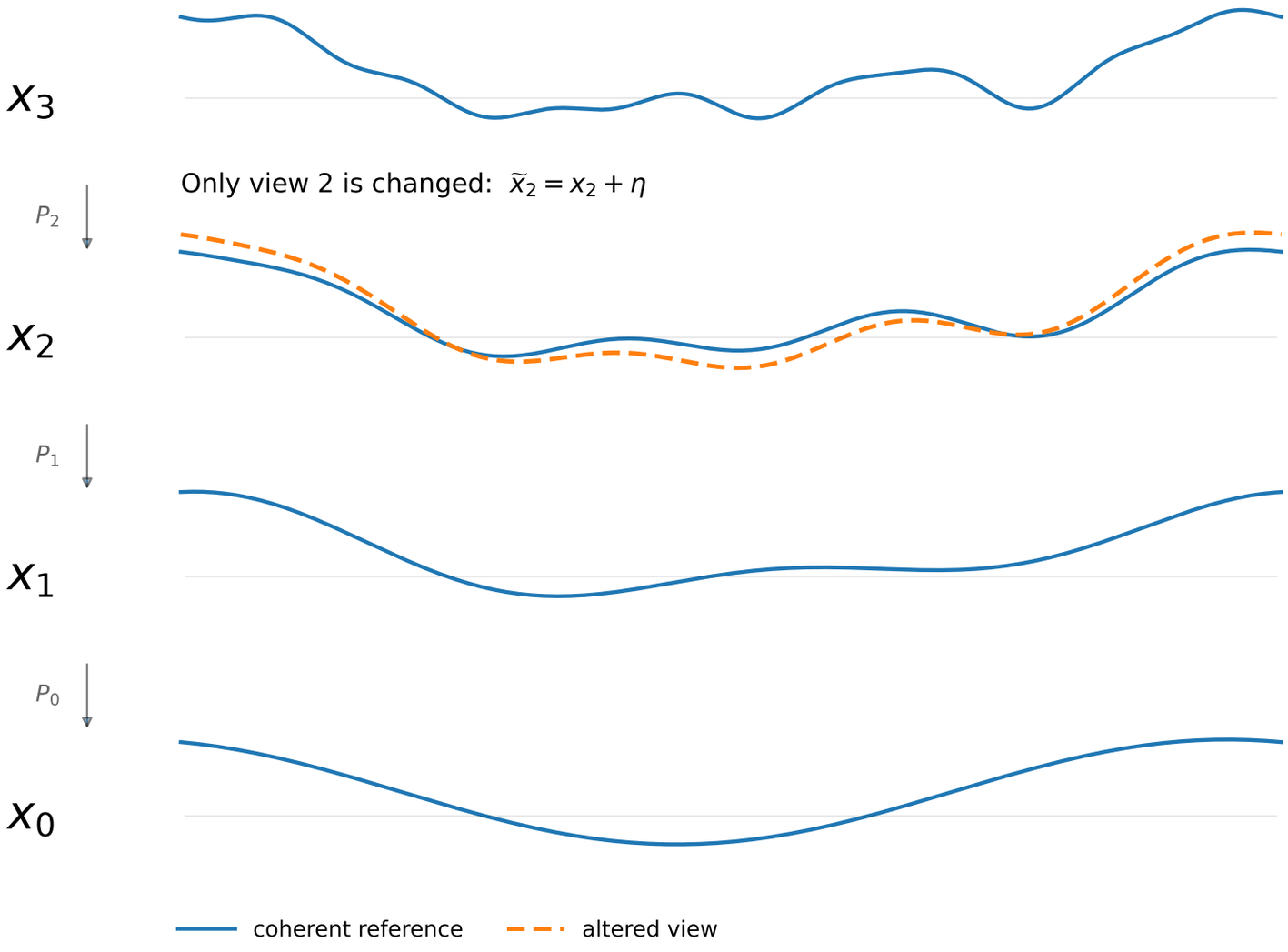
Finite illustration of the realization construction. Geometry and identities are reproducible in Python.

Detail is not inconsistency

Change one view. The compatibility differential records exactly where agreement fails.

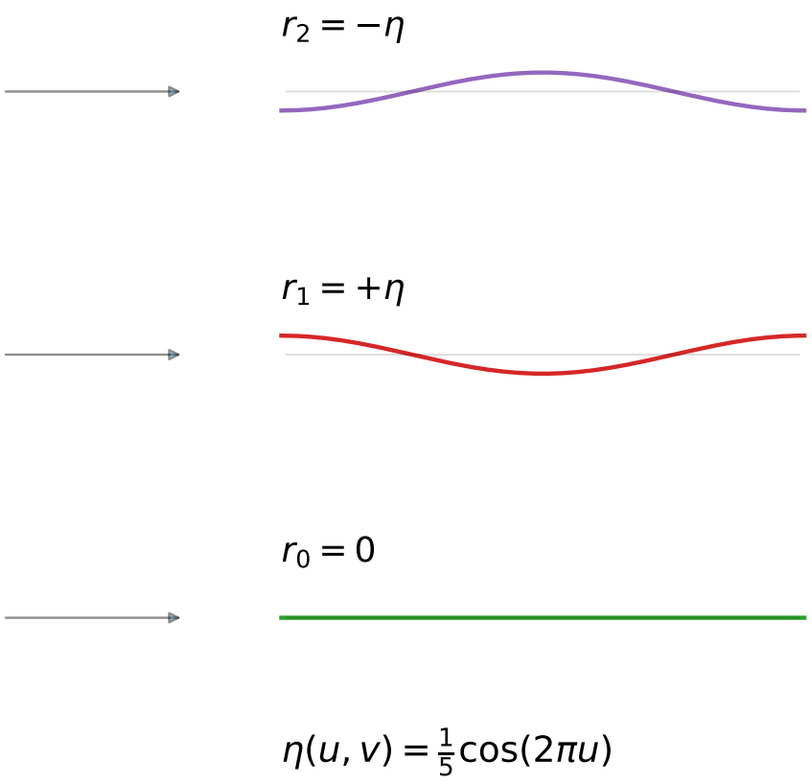
VIEWS OF THE SAME FAMILY

Signal slice at $v = 3/16$



TYPED DEFECTS

$r_n = P_n \widetilde{x}_{n+1} - \widetilde{x}_n$



RETAINED DETAIL

$$d_n = x_{n+1} - J_n x_n \in \ker P_n$$

May be nonzero even when the family is perfectly coherent.

COMPATIBILITY DEFECT

$$\delta\widetilde{x} = (0, \eta, -\eta) \neq 0$$

Subtracting η from view 2 repairs this family exactly.

This defect is repairable. A nonzero residual alone is not evidence of a nonzero extension obstruction.